

Understanding Memory in C

A technical paper by Kent Tonino

Preface

In C, you have direct control over memory. In Python or Go, memory is managed for you, variables appear when you need them and disappear when you don't, without you thinking about it. In C, you are responsible. You decide where data lives, how long it lasts, and when to release it.

For me, that responsibility is not a burden, it is a capability. Understanding memory in C helps you understand what is going on under the hood. With this out of curiosity paper, it covers memory from the ground up, on how a C program's memory is organised, how the stack and heap work, what pointers actually are, why structs have unexpected sizes, what goes wrong when memory is misused, and how to find those bugs.